
Decode Microkernel Consolidation for Hybrid Reasoner Serving

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Abstract

This draft reports the current state of Decode Microkernel Consolidation (DMC), a positive-method pivot produced after a preregistered Hybrid-Kernel boundary-fusion gate failed. DMC targets repeated launch-heavy decode micro-operations rather than attention/SSM boundaries. The current evidence is a preregistered Phase 1 replay gate: a trace-derived packed Triton replay path reduces GPU-side replay latency while preserving numerical equivalence on three primary, three same-family, and three cross-family schedules. This is not yet an end-to-end serving speedup claim. The next preregistered gate must show that the same method improves real vLLM/serving decode latency on frozen prompt slices.

1 Introduction

Hybrid and mixture-of-experts language models expose many small decode-time GPU operations. A failed HybridKernel profiler gate found no separable attention/SSM boundary-local overhead, but the same server-side Nsight artifacts showed dense repeated micro-operation families during decode. DMC asks whether those repeated operations can be consolidated into fewer GPU launches without changing outputs.

The submission status is deliberately conservative. Phase 1 is a replay implementation gate, not a serving result. The paper is not camera-ready until Phase 2 shows an end-to-end serving gain with frozen prompts, model outputs, launch audits, and paired uncertainty.

2 Related Work

This interim draft does not make external novelty claims beyond the repository audit. The camera-ready candidate must include a hand-verified citation audit covering serving launch overhead, compiler fusion, CUDA Graphs, Triton kernels, and hybrid/MoE decode systems. Until that audit is complete, the only cited evidence is internal artifact evidence listed in the reviewer pack.

3 Method

DMC builds a fixed replay schedule from committed server-side Nsight Systems `cuda_gpu_kern_sum` logs in the killed HybridKernel packet. For each source row, the runner maps kernel-family weights into a packed gating/routing/state-update replay schedule. The baseline executes repeated PyTorch CUDA micro-operations; the consolidated path executes one Triton packed gating replay kernel. The checker verifies CUDA-event timing samples, launch audit artifacts from `torch.profiler`, saved output tensors, source artifact hashes, fixed packet identity, and numerical equivalence.

The claim boundary is strict: DMC Phase 1 validates a replay mechanism. It does not validate model quality, vLLM integration, attention/SSM boundary fusion, or end-to-end serving acceleration.

Role	Rows	Median latency reduction	95% CI low	Max abs error
Primary hybrid	3	0.9650953811812909	0.9618806519742926	7.152557373046875e-07
Same-family control	3	0.9657258621103347	0.9656092336667471	5.960464477539062e-07
Cross-family falsification	3	0.9655504614632344	0.9654552838317912	5.960464477539062e-07

Table 1: Phase 1 replay-gate metrics from the checker output cited in Section 4. The preregistered thresholds were median latency reduction at least 0.08 for primary and same-family roles, at least 0.05 for cross-family roles, bootstrap 95% CI lower bound greater than 0 for every role, launch reduction at least 0.25 in every row, max absolute error at most $1e-2$, and max relative error at most $1e-2$.

4 Experiments

Phase 1 used the fixed PASS packet experimental/decode_microkernel/phase1/results/dmc_phase1_20260508T000525Z. The gate was preregistered in experimental/decode_microkernel/phase1/preregister_dmc_phase1.md. The checker output is experimental/decode_microkernel/phase1/results/dmc_phase1_20260508T000525Z/checker_result.json.

Each row used at least 100 warmup iterations and 1000 measured CUDA-event iterations. All nine rows were checked against saved output tensors and per-row launch-audit artifacts.

5 Results

The checker returned PASS_DMC_PHASE1_CONSOLIDATED_REPLAY. The minimum row-level launch reduction was 0.9916666666666667. The maximum relative error was 8.915998250813573e-07.

6 Discussion

The Phase 1 result is a useful positive signal because it shows that a trace-conditioned packed replay can consolidate launch-heavy decode micro-operations with exact artifact checks. It is not sufficient for a COLM method paper by itself. The replay baseline intentionally exposes launch overhead; an optimized serving stack may already remove some of that overhead. The next gate therefore measures real serving rows and requires generated-token matches, end-to-end decode latency reduction, tokens/sec gain, and launch audits.

7 Limitations

Phase 1 has only nine replay rows and three rows per role. Bootstrap intervals over three rows should not be treated as paper-strength uncertainty. The workload is trace-derived but synthetic; it does not exercise full model generation, KV-cache behavior, batching, sampling, prefill/decode interaction, or existing vLLM compiler and CUDA Graph optimizations. Same-family evidence includes one row whose Phase 0 SQLite summary was empty, although Phase 1 derives that schedule from the committed server-side Nsight log and records that provenance explicitly. Phase 2 must resolve these limitations before camera-ready status.

8 Reproducibility Statement

The current Phase 1 packet was produced on commit 86fd5b027ea6259cbba9be60a81427efa2d525f5 on an NVIDIA RTX PRO 6000 Blackwell Server Edition GPU. Runtime was less than 0.01 GPU-hours for the canonical replay packet. Reproduction uses:

HF_HOME=/workspace/hf_cache TRITON_CACHE_DIR=/workspace/.triton_cache \

```
./venv_gpu/bin/python experimental/decode_microkernel/phase1/run_dmc_phase1.py \
--run-id dmc_phase1_20260508T000525Z
```

```
./venv_gpu/bin/python experimental/decode_microkernel/phase1/check_dmc_phase1.py \
experimental/decode_microkernel/phase1/results/dmc_phase1_20260508T000525Z
```

All claims in this draft cite experimental/decode_microkernel/phase1/results/dmc_phase1_20260508T000525Z.

9 Ethics Statement

This work is a systems optimization study. The current artifacts do not involve human-subject data, user data, or deployment claims. The main ethical risk is overclaiming a replay microbenchmark as a serving speedup; the paper explicitly forbids that interpretation until Phase 2 passes.